ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 39 per cent	64/164 x 100 = 39.
3	**(a)** Y	Immiscible liquids.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 16 per cent	20+60 = 80 g in 500 g.
6	**(b)** a solid dissolves in a solid	Metals.
7	**(c)** A is true but R is false.	A is true; R is false: powder has a larger surface area.
8	**(b)** 25 per cent	100/400.
9	**(c)** dissolved solids	Residue.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Percentage = 37.5 per cent (1); Water = 62.5 g (1).	1+1
12	Dissolved air is driven out on boiling (1); Little gas redissolves on cooling (1).	1+1
13	The solution is more concentrated (1); More sugar per unit mass of solution (1).	1+1
14	Salt = 45 g (1); Water = 705 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	NaCl: 4 g undissolved (1); KNO3: 8 g undissolved (1); NaCl is more soluble (1).	1+1+1
16	Dissolve impure crystals in hot water (1); Filter the hot solution (1); Cool slowly to get pure crystals (1).	1+1+1
17	Rivers carry dissolved minerals into the sea (1); Evaporation removes only water (1); Salts accumulate over time (1).	1+1+1
	OR: Sugar = 60 g (1); New total = 60/0.12 = 500 g (1); Water added = 300 g (1).	1+1+1
18	Suspension settles on standing; colloid doesn't (1); Suspension is filtered out; colloid passes through filter paper (1); Both show Tyndall effect (1).	1+1+1
	OR: It spreads as tiny particles (1); Mass is conserved (1); It can be recovered by evaporation (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At 70: 150 g solution = 50 g salt + 100 g water (1); At 20 the 100 g water holds 20 g (1); Crystals = 50 - 20 (1); = 30 g (1); Percentage of solution at 20: 20/120 = 16.7 per cent (1).	1+1+1+1+1
	OR: Boil solution in a flask (1); Vapour passes through a condenser (1); Pure water collects (1); Salt remains (1); No dissolved impurities (1).	1+1+1+1+1
20	Equal volume of water in each beaker (1); Add 1 g portions with stirring until saturation (1); Same temperature and stirring (1); Sugar likely most soluble; chalk insoluble (1); Record in a table (1).	1+1+1+1+1
	OR: Alum settles suspended particles (1); Filtration removes insoluble particles (1); Boiling/UV kills germs (1); Softener removes dissolved salts (1); Dissolved vs suspended impurities need different methods (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 300 g	40/0.1 = 400 total; add 300.
2	**(c)** copper sulphate	Blue Cu2+.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** 60 g water and 25 g crystals	At 20: 15 g dissolves in 60 g, so 25 g crystallises.
5	**(a)** remains the same	Mass is conserved.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(d)** salt	Dissolved solute.
8	**(a)** sugars in water	Concentrated sugar solution.
9	**(c)** A is true but R is false.	A is true; R is false: milk is a colloid.
10	**(c)** coarse crystals in cold water without stirring	All factors slow.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gas is dissolved under pressure (1); Pressure release makes gas escape (1).	1+1
12	The substance in smaller quantity is the solute by convention (1); Water can act as a solute in such cases (1).	1+1
13	Total = 24/0.08 = 300 g (1); Solvent = 276 g (1).	1+1
14	Both dissolve in water (1); They pass through filter paper (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heating reduces gas solubility (1); Pressure builds up (1); May burst or spray (1).	1+1+1
16	Gas particles gain energy (1); Escape from solution (1); Less gas stays dissolved (1).	1+1+1
17	Heat for camphor by sublimation (1); Dissolve the rest and filter sand (1); Evaporate for salt (1).	1+1+1
	OR: Heat tap water gently in a flask (1); Tiny bubbles appear before boiling (1); Dissolved gases escape (1).	1+1+1
18	Solute particles separate and spread between solvent particles (1); Sand particles remain as lumps (1); Solution is uniform (1).	1+1+1
	OR: First: 15 per cent (1); Second: 7.5 per cent (1); First is twice as concentrated (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Light scattered by colloids (1); True solution does not scatter (1); Suspension scatters and settles (1); Beam of sunlight through forest mist or headlights in fog (1); Particle size decides (1).	1+1+1+1+1
	OR: Nature (1); Size (1); Filtration and settling (1); Tyndall effect (1); Examples (1).	1+1+1+1+1
20	Solute particles separate and spread (1); Dilution adds solvent particles (1); Saturation: no more can fit (1); Higher temperature: more motion, more solubility (1); Gases leave at high temperature (1).	1+1+1+1+1
	OR: Salt = 20 g (1); Water = 480 g (1); Dissolve with stirring (1); Add a little more salt: it dissolves (1); So solution is unsaturated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** carbon dioxide in water	Soda-acid type.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Large particles stay on the filter.
3	**(a)** greater than x	Pressure increases gas solubility.
4	**(b)** solubility depends on solute and solvent	Selectivity.
5	**(c)** A is true but R is false.	A is true; R is false: solute mass is unchanged and solvent increases.
6	**(a)** about 18 per cent	20/110 x 100 = 18.
7	**(a)** high gas pressure from reduced gas solubility	Gas comes out.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Like dissolves like.
9	**(a)** equilibrium between dissolved and undissolved salt	Saturated.
10	**(a)** distillation	Boiling points differ.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dilute solution is still unsaturated at the lower temperature (1); Saturated solution exceeds solubility (1).	1+1
12	Cold water contains more dissolved oxygen (1); Solubility of gas decreases with temperature (1).	1+1
13	Dissolving: salt in water, recoverable (1); Reacting: iron with acid, new substance (1).	1+1
14	Safe concentration is required (1); Sterile to avoid infection (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	More evaporation (1); Water leaves; salt crystallises (1); Sunlight and wind help (1).	1+1+1
16	Add more solute (1); Evaporate solvent (1); Use a smaller volume of solvent (1).	1+1+1
17	Paint: solvent evaporates leaving pigment (1); Perfume: alcohol carries fragrance (1); Medicine: water or alcohol carries drug (1).	1+1+1
	OR: Stains may be insoluble in water (1); Acetone for nail polish (1); Petrol or turpentine for grease or paint (1).	1+1+1
18	Darker blue means more concentrated (1); Lighter blue means more dilute (1); Colour intensity relates to amount of solute (1).	1+1+1
	OR: Water evaporates (1); Concentration of sugar rises (1); Syrup becomes viscous (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Oxygen solubility is less in warm water (1); Fish need dissolved oxygen (1); Organic pollution consumes oxygen (1); Algal bloom depletes oxygen (1); Fish die (1).	1+1+1+1+1
	OR: Sugar solubility increases with temperature (1); Hot saturated solution cools (1); Excess sugar crystallises (1); String or stick provides surface (1); Large crystals form slowly (1).	1+1+1+1+1
20	Salt = 10 + 45 = 55 g (1); Total = 500 g (1); Percentage = 11 per cent (1); Water = 445 g (1); Concentration lies between 5 and 15 (1).	1+1+1+1+1
	OR: Final mass of solution = 120 g (1); Percentage = 20/120 x 100 = 16.7 per cent (1); Mass of dry residue = 20 g (1); Mass is conserved in dissolving (1); Evaporation removes only the solvent (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.